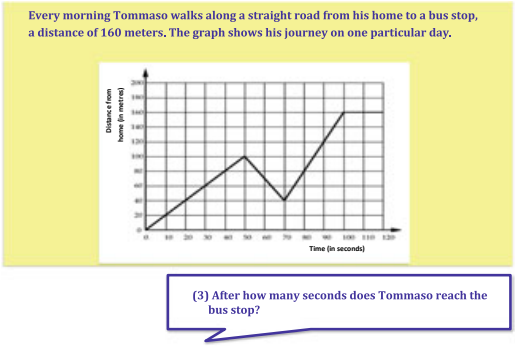
Table 1 Categories of polls resulting from our design experiments

Category of poll	Corresponding aims	Example from our design experiments
(1) Polls on specific mathematical content: these polls ask to choose the correct answer to a problem or to a specific question	– To highlight students' understanding of specific topics developed during the lesson – To make students discuss the reasons linked to the choice of the correct answers (providing feedback about the task) – To promote a discussion on the solving strategies, in order to enable students to share and compare them (providing feedback about the processing of the task)	Every morning Tommaso walks a straight road from his home to a bus stop, a distance of 160 m. The graph shows his journey on one particular day  After how many seconds does Tommaso reach the bus stop? (a) After 120 s (b) After 50 + 70 + 100 + 120 s, that is after 340 s (c) After 100 s (d) After 50 s

(continued)

Table 1 (continued)

Category of poll	Corresponding aims	Example from our design experiments
(2) Polls on argumentation: these polls ask students to compare different justifications about the answers to a given problem	– To promote a discussion at a meta-mathematical level, focused on the way the answer is justified. Thanks to this kind of discussion, students are led to identify the criteria to assess the answers. Hence, students receive and provide feedback on the processing of the task and feedback about self-regulation	See Sect. 7
(3) Polls on metacognitive aspects: these polls are focused on the difficulties students meet when facing specific kind of tasks or on the best strategies to be used to face specific tasks	– To promote metacognitive reflections that could help students in identifying the available tools to face similar tasks in the future and in becoming aware of how to monitor themselves while facing this kind of task (providing *f*eedback about self-regulation)	See Sect. 8
(4) Polls on affective aspects: these polls are focused on students' feelings when facing a specific kind of task or when a particular methodology is adopted during the lessons	– To bring to the fore the affective dimension, supporting students in becoming aware of their way of posing themselves during this kind of activity (providing feedback about self-regulation)	How did you feel when your answer was displayed on the interactive whiteboard? A) Uneasy B) Happy C) Calm D) Worried